

SRI VIJAY VIDYALAYA COLLEGE OF ARTS AND SCIENCE

PG & Research Department of Computer Science

EDU GENIE

Google Gemini Powered Learning Assistant

TEAM

G. DHANALAKSHMI

Team Leader

MEMBERS

- P. KAVIYA
- S. KAVIYA
- G. APSARA
- S. DHATCHANA

Google Gemini

Why Google Gemini?

Next-generation multimodal AI built for speed, accuracy, and real-world understanding



Multimodal understanding

Text • Image • Voice • Video



Super Fast & Accurate

Industry-leading response time with precise, reliable results



Understands Handwritten Notes

Reads, analyzes, and summarizes handwritten content with ease



Google Search integration

Real-time access to the latest information via Google Search



Problem Statement in Education

Key challenges holding learners back today



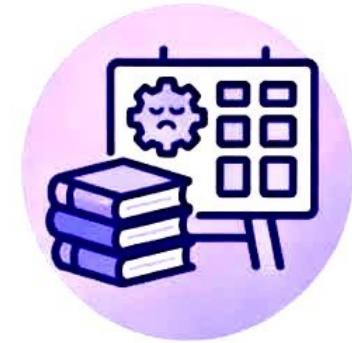
Students Confused

- Overwhelmed by endless information
- Lack of clarity on what to study
- Struggle to stay motivated



Expensive Tutors

- High hourly fees unaffordable for many
- Limited access to quality help
- Inconsistent learning experience



Boring One-Size Teaching

- Generic lessons for all students
- No personalization or engagement
- Passive learning, low retention

The result: Disengaged learners • Lower outcomes • Growing learning gap



Edu Genie

AI-Powered Learning Assistant

Key Features

Instant Doubt Solving



Instant Doubt Solving

Upload a photo of your question and get step-by-step solution instantly



Personalized Learning Path

Tailored study roadmap based on your progress & learning pace



Smart Quiz Generator

Auto-generated quizzes to test knowledge & reinforce learning



Voice Support in Tamil & English

Ask and learn hands-free with natural voice support in Tamil & English



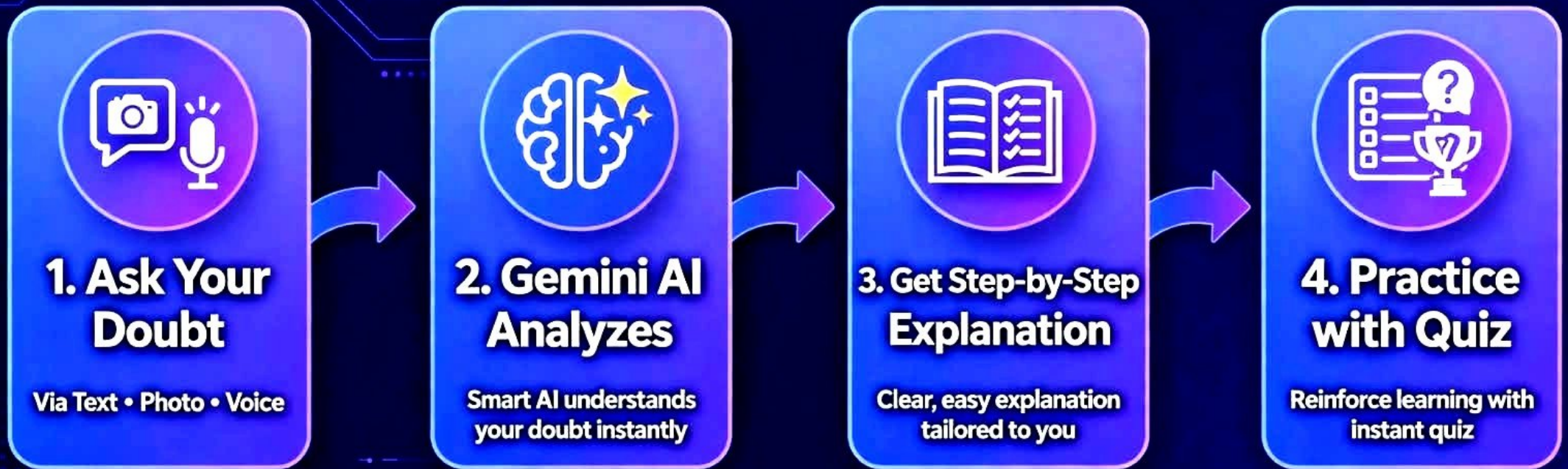
Chapter Summarizer

Get concise chapter summaries to revise faster & retain better

✦ Empowering Students • Learn Faster • Understand Better ✦

HOW IT WORKS

 **Edu Genie • 4 Simple Steps to Learn & Grow**



Learn. Practice. Succeed.

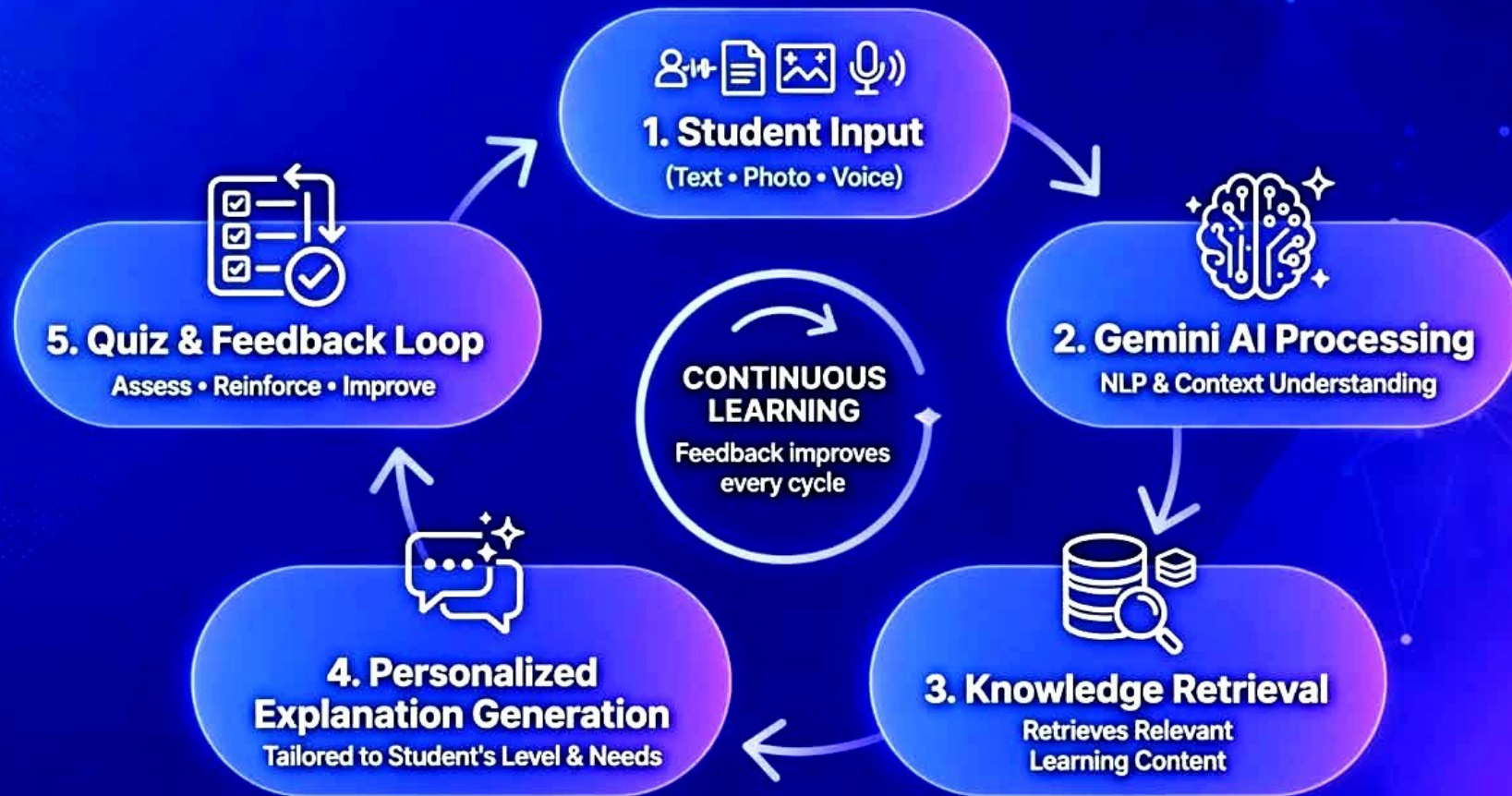
Empowering Tamil Nadu Students



Edu Genie

WORKING PROCESS

HOW IT WORKS





Edu Genie
Intelligent Learning Platform

MODULES — SYSTEM MODULES



1. User Authentication Module

Secure login, role-based access, and user profile management



2. Doubt Solving Module with Gemini AI

Real-time doubt resolution powered by Gemini AI for instant answers



3. Personalized Learning Module

Adaptive learning paths tailored to individual progress and goals



4. Quiz and Assessment Module

Interactive quizzes, assessments, and performance tracking



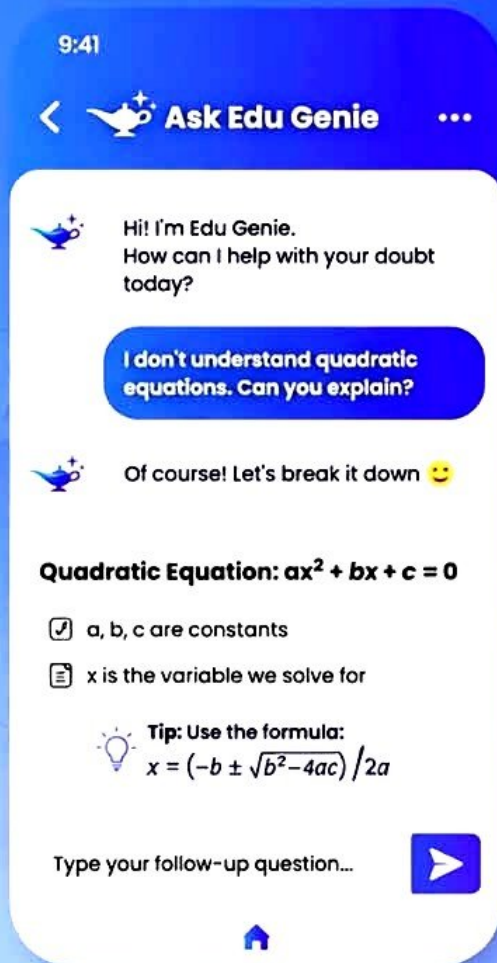
5. Admin and Analytics Module

Dashboard insights, user analytics, and system management

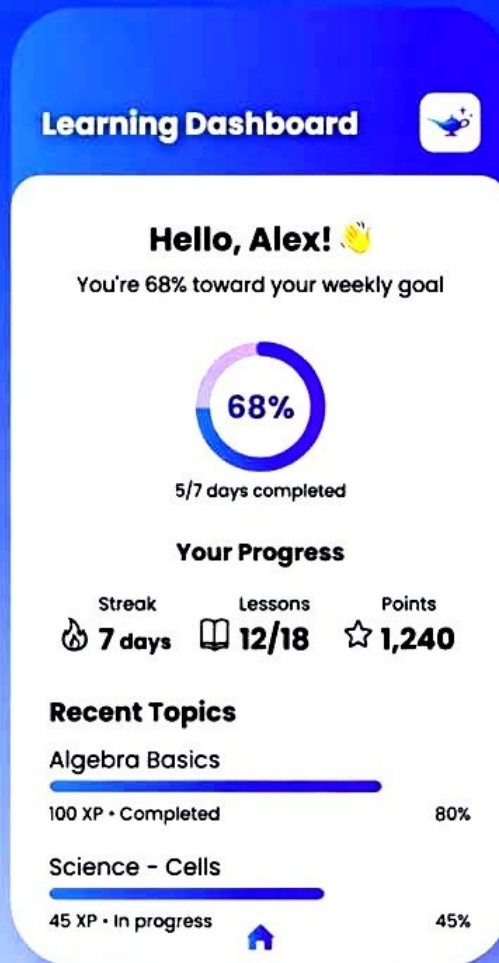
Scalable • Secure • AI-Powered • Data-Driven

USER INTERFACE – UI DESIGN

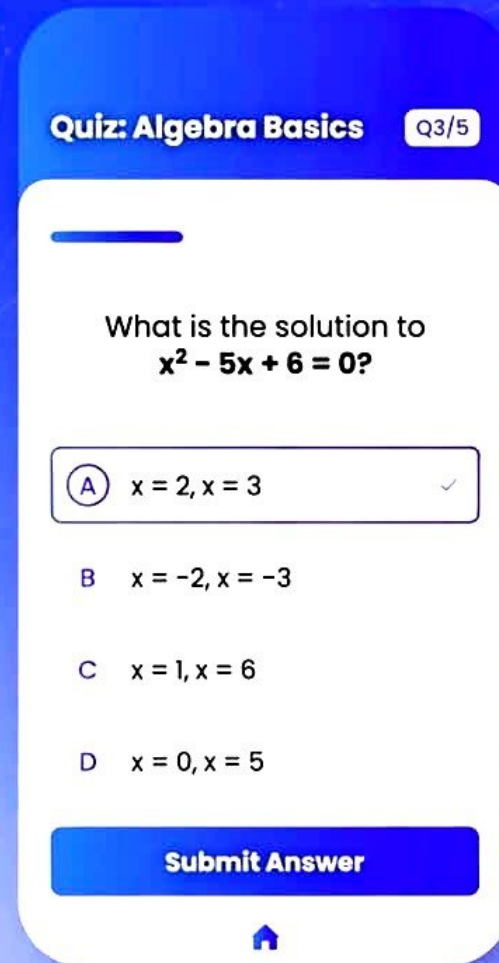
Edu Genie • Modern Mobile App UI Design



Chat • Doubt Solving



Dashboard • Learning Progress



Quiz • Knowledge Check

AI-Powered Learning Platform



Edu Genie •

TECHNOLOGIES USED

TECH STACK

FRONTEND



React • Tailwind

Interactive UI • Responsive Design

BACKEND



Node.js • Express

REST API • Server-side Logic

AI & ML



Google Gemini 1.5 Pro • NLP

Natural Language Processing • AI Tutoring

DATABASE



Firebase • Firestore

Real-time • Cloud Database

TOOLS



VS Code • Git • Figma

Development • Version Control • Design

Powering adaptive, intelligent education through modern technologies

APPLICATIONS

WHERE IT CAN BE USED



School Education

K-12 classrooms, homework support, interactive lessons



Competitive Exam Prep

JEE, NEET, IELTS, standardized test practice



College Learning

University courses, research support, study assistance



Rural Education

Offline access, local language support, community learning



Skill Development

Coding, communication, professional skills training



Distance Learning

Remote classes, virtual tutoring, anytime access



Edu Genie
Personalized AI Tutoring Platform

Empowering learning everywhere • AI-driven • Accessible



Edu Genie
AI-Powered Learning Companion

ADVANTAGES



24/7 Instant Doubt Solving

Get answers to questions anytime, anywhere with immediate AI support



Personalized Learning Path

Tailored lessons that adapt to each student's pace and learning style



Affordable and Accessible

Quality education at low cost, available on any device for all learners



Photo • Voice • Text Input

Ask questions via photo, voice command, or text for flexible learning



Interactive Quiz Practice

Reinforce learning with engaging quizzes and instant feedback

Empowering students • Smarter learning • Better results

DISADVANTAGES

CHALLENGES



Internet Dependency

Requires stable internet connection; limited functionality offline



Limited Human Emotional Support

Lacks empathy and personal reassurance that human tutors provide



Potential AI Hallucination

May generate inaccurate or misleading information if not verified



Over-reliance Risk

Students may become too dependent and reduce independent learning



Privacy Concerns

Data security and user privacy require strict protection protocols



Edu Genie
Empowering Learning with AI

FUTURE ENHANCEMENTS

Expanding Edu Genie to empower learners everywhere



Voice Assistant Integration

Hands-free learning via voice commands & Q&A for accessible, interactive support



AR • VR Learning

Immersive augmented & virtual reality experiences to visualize concepts in 3D



Multi-Language Support • Tamil etc.

Expand to Tamil, Hindi, Malayalam, and more — breaking language barriers for all learners



AI Career Guidance

Personalized career recommendations and learning paths based on skills & interests



Gamification & Leaderboards

Earn points, badges & compete on leaderboards to boost motivation and engagement



Offline Mode

Access lessons and materials without internet — sync progress when reconnected



THANK YOU